Exercise 2A

2) (a) scalar (c) vector (e) vector
(b) vector (d) scalar (f) speed

6) Displacement is the shortest distance from the initial to the final position of the body, is the magnitude of displacement and its direction is from the initial position to the final position. The unit of displacement is metre (m).

7)

Distance	Displacement
⇒ It is the length of the path traversed by the object in a certain time.	It is the distance travelled by the object in a specified direction in a certain time.
⇒ It is a scalar quantity	⇒ It a vector quantity.
⇒ It depends on the path followed by the object.	It does not depend on the path followed by the object.
⇒ It is always positive	It can be + or −
⇒ ~~It can be more than or equal to the distance but can never be greater than the~~	
⇒ It can be more or equal to the magnitude of displacement.	Its magnitude can be less than or equal to the distance, but can never be greater than the distance
⇒ It may not be zero even if displacement is zero, but it can not be zero if displacement ≠ 0	It is zero if the distance is zero, but it can be 0 if distance is not zero.

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10) The velocity of a body is the distance travelled per second by the body in a specified direction. The unit of velocity is m/s.

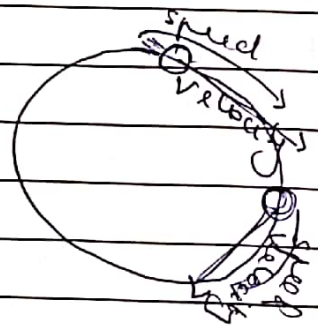
13) Velocity gives us the direction

14) When the velocity is constant.

15) ⇒

Variable velocity	~~Uniform Velocity~~ Uniform Velocity
The velocity of the object might differ in different instances.	The Velocity of the object never changes at any given instance.

17) When an object in a circular path with a constant speed, the velocity of the object is variable because the direction of motion changes constantly.



Here the speed remains constant, but as the direction continues to change, the velocity is variable.

19) Acceleration is the rate of change of velocity over time. The S.I unit of acceleration is "m/s^2" or "$m\,s^{-2}$".

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21)

Uniform Acceleration	Variable Acceleration
In uniform acceleration there is an equal change in ~~equal time intervals~~ velocity over equal time intervals	In variable acceleration there is an unequal change in velocity over equal time intervals.

22) Retardation is the decrease of velocity over a time period. It can also be called as negative acceleration or deacceleration. The S.I unit of deacceleration is "m/s^2".

23) The car was moving with a constant velocity initially but then it start to deaccelerate.

27) No. Because the effect of gravity on an object depends where it is. The value of g is the greatest near the poles and least near the equator.

MCQS

1) (d) 2) (d) 3) (d) 4) (d) 5) (d)

Numerical :-

1) 72 km/h = 72000 m/h

$\Rightarrow \frac{72000 \text{ m}}{3600 \text{ s}}$

$\Rightarrow 20 \text{ m/s}$

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3) (a) $1\,km/h = \frac{1000\,m}{3600\,s} = 0.278\,m/s$

(b) $18\,km/min = \frac{18000\,m}{60\,s} = 300\,m/s$

4) (i) $10\,m/s = 10\,m/s$

(ii) $1\,km/min = \frac{1000\,m}{60\,s}\ 16.6\,m/s$

(iii) $18\,km/h = \frac{1800\,m}{3600\,s}$

$= 5\,m/s$

6) 100 / 6 = 16.6

40 / 36 = 4

So $5\,m/s < 10\,m/s < 16.6\,m/s$

∴ Answer = ~~5 m/s, 10 m~~

$18\,km/h < 10\,m/s < 1\,km/min$

6) Let the time required for the first 30 km be t_1 and the time required ~~at the~~ for the other half be t_2.

$t_1 = \frac{30}{60} = 0.5\,hrs = 30\,min$

$t_2 = \frac{30}{40} = 0.75\,hrs = 45\,min$

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(i) Total time $= t_1 + t_2 = 30\,min + 45\,min$
$= 75\,min$

(ii)

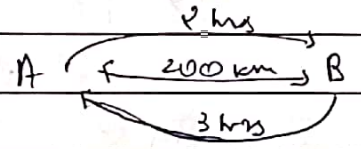
$$\text{avg speed} = \frac{\text{total distance}}{\text{total time}}$$

$$\Rightarrow \frac{30 + 30\ km}{1.25\ hrs}$$

$$\Rightarrow \frac{60\ km}{1.25\ hrs}$$

$$\Rightarrow 48\ km/hr$$

7)



(i) $$\text{avg speed} = \frac{\text{total distance}}{\text{total time}}$$

$$\Rightarrow \frac{200\,km + 200\,km}{3\ hrs + 2\ hrs}$$

$$\Rightarrow \frac{400\ km}{5\ hrs}$$

$$= 80\ km/h$$

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$v^2 = u^2 + 2aS$

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(ii) Zero because displacement = 0.

8)

(i) $\frac{1000\ m}{100\ s} = 10\ m/s$

(ii) 10 m/s due east

10) 0.05 hrs = 180 sec

Sol

$a = \frac{v-u}{t}$

$= \frac{180\ m/s - 0\ m/s}{180\ s}$

$= \frac{180\ m/s}{180\ s}$

$= 1\ m/s^2$

12) $a = \frac{v-u}{t}$

$= \frac{0-18}{2}$

$=$

18 km/h = $\frac{18000}{3600}$ $\frac{180}{36}$ 5 m/s

$a = \frac{0-5}{2}$

$= -2.5\ m/s$

∴ Retardation = 2.5 m/s

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14)

$$\cancel{t = \frac{v-u}{a}}$$

$$\cancel{t = \frac{0-20}{2}}$$

$$\cancel{= 10s}$$

$$\cancel{5 = \frac{v-20}{2}}$$

14) $V = at$

$= -2 \times 5$

$= 10 m/s$

∴ The final velocity would be 10 m/s

16) (i) $\frac{1800m}{360s} = 5 m/s$

(ii) $a = \frac{0 - 5}{5}$

$= -1 m/s^2$

∴ retardation $= 1 m/s^2$

(iii)

$V = a \times t$

$= -1 \times 3$

$= 3 m/s$

∴ Velocity after 3 seconds would be 3 m/s

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